Attrition Prevention Suggestions

HR Analytics — Employee Attrition Prediction

1. Key Attrition Drivers Identified

SHAP (SHapley Additive exPlanations) analysis on the Logistic Regression model identified the following factors as the strongest predictors of employee attrition, ranked by impact:

- Overtime — the single strongest driver; employees working overtime attrite at much higher rates.
- Frequent business travel — employees who travel frequently show elevated attrition risk.
- Job role (Laboratory Technician) — this role shows higher-than-average attrition.
- Total working years — attrition risk varies by career stage, with early/mid-career employees more likely to leave.
- Environment satisfaction — lower satisfaction with the work environment strongly correlates with leaving.
- Years since last promotion, number of companies worked, and job satisfaction were also meaningful contributing factors.

2. Recommended Prevention Strategies

2.1 Manage Overtime and Workload

- Audit teams/roles with high overtime frequency and assess whether workload should be redistributed or headcount increased.
- Introduce workload monitoring dashboards for managers to flag sustained overtime early.
- Offer compensatory time off or overtime pay adjustments to reduce burnout-driven exits.

2.2 Reassess Travel Demands

- Review roles requiring frequent travel and evaluate whether virtual alternatives can reduce travel load.
- Introduce travel frequency caps or rotation policies so the burden doesn't fall on the same employees repeatedly.
- Provide additional travel-related benefits (e.g., recovery days, travel allowances) for high-travel roles.

2.3 Target High-Risk Roles

- Conduct focused stay interviews with Laboratory Technicians and other high-risk roles to understand role-specific pain points.
- Benchmark compensation and career growth paths for these roles against market standards.

2.4 Support Career Growth

- Create clearer promotion timelines and communicate growth paths, especially for employees several years since their last promotion.
- Introduce mentorship or internal mobility programs for early/mid-career employees identified as flight risks.

2.5 Improve Workplace Environment & Satisfaction

- Run regular pulse surveys to track environment and job satisfaction, using survey scores as leading indicators.

- Invest in manager training, since manager relationships strongly influence day-to-day satisfaction.
- Improve physical/remote work environment factors flagged in employee feedback.

2.6 Operationalize the Predictive Model

- Use the trained model to score current employees monthly or quarterly, flagging those with high predicted attrition risk.
- Route flagged employees to HR business partners for proactive check-ins before resignation risk materializes.
- Retrain the model periodically as new attrition data becomes available to maintain accuracy.

3. Expected Impact

By addressing the top drivers identified — overtime, travel burden, role-specific dissatisfaction, and stalled career progression — the organization can shift from a reactive exit-interview approach to a proactive, data-driven retention strategy. Even modest reductions in attrition among high-risk segments can meaningfully lower recruitment and training costs while preserving institutional knowledge.